Structural Certificates for Prediction Depth

Frozen and transported control in one-sided cellular automata

Research note with Lean 4 proofs

Abstract. We classify the prediction depth of one-sided cellular automata whose frozen control words gate a stay/hop data update over $\mathbb{F}_2$. The input is the explicit initial cone and the output is the data bit at its apex. For each fixed rule, a single logspace-uniform family of polynomial-size De Morgan circuits realizes the classified depth upper bound. The variable-guard regime has depth $\Theta(\log(T+1))$, and homogeneous Pascal prediction has depth within a factor of three of the number of ones in the binary expansion of T. We also analyze a moving-control model in which a single data path accumulates Boolean guards. At positive times, equal control and data velocities give depth one, whereas unequal velocities give exact depth $\lceil \log_2(T+1) \rceil$. The finite statements follow from Lean 4 proofs of the dynamics, circuit correctness, and the fan-in lower bounds.

1 Prediction problem and circuit model

A local rule on configurations over $\mathbb{Z}$ is iterated T times. We ask for one data bit, at site zero. Each model in the main results has a one-sided radius-one update, so its initial window consists of sites $0, \dots, T$. For the frozen model the input contains $q(T+1)$ control bits followed by $T+1$ data bits. The control width q and the two guard circuits are fixed independently of T. The legal word lengths are $n = (q+1)(T+1)$ and determine T uniquely. The language consists of the legal words whose predicted bit is one, with all other lengths rejected.

Our upper bounds use Boolean AND and OR gates of fan-in two, NOT gates, and constants. An input wire has depth zero. Gates can share outputs and connect arbitrary sites. The per-time lower bounds apply to any Boolean gates of fan-in at most two in this wiring model. For a fixed rule, write $D(T)$ for the minimum De Morgan circuit depth on the legal window at time T and write $\ell(m) = \lceil \log_2 m \rceil$ for $m \geq 1$. Uniform circuit families are indexed by input length. A deterministic logspace transducer receives the length in unary and writes the complete circuit code, including its gates, wiring, and output. We use logspace uniformity throughout the statements about NC.

A coordinate is **essential** if two assignments that differ only at that coordinate give different outputs. The backwards cone of a depth-d output with fan-in at most b contains at most b^d input coordinates. This follows by induction on depth, since a gate has at most b predecessor cones. It remains valid in a shared circuit because counting a repeated predecessor more than once can only increase the bound. Thus m essential coordinates force $d \geq \lceil \log_b m \rceil$ when $b \geq 2$ and $m \geq 1$.

2 Frozen controls

2.1 The iterate of the local rule

Let $a_i \in \{0,1\}^q$ be a control word, $x_i \in \mathbb{F}_2$ a data bit, and $s, h : \{0,1\}^q \to \{0,1\}$ the stay and hop guards. The synchronous update is

$$a_i' = a_i, \quad x_i' = s(a_i)x_i + h(a_i)x_{i+1}, \tag{1}$$

where the data arithmetic is in $\mathbb{F}_2$. A backwards path ending at site k has k hops and $T-k$ stays. Put

$$e_k = \prod_{j=0}^{k-1} h(a_j), \quad w_k = \sum_{j=0}^{k} s(a_j), \tag{2}$$

where the second sum is in the natural numbers. For natural w, r, write $M(w, r)$ for the number of weak compositions of r into w parts. For $w > 0$, it is $\binom{w+r-1}{r}$, and $M(0, r)$ is one at $r = 0$ and zero otherwise.

Theorem 1 (iterate formula). For every initial configuration and time T, the data bit at site zero is

$$x_0(T) = \sum_{k=0}^{T} e_k (M(w_k, T-k) \bmod 2) x_k. \tag{3}$$

The same identity holds at any site after translating the indices.

Each hop must be enabled, contributing the factor e_k. A path visits the sites 0 through k in order. The number of stays allocated to each enabled site determines the path, including possible stays after the last hop. The Lean proof in `iterate_data` derives the recurrence for these coefficients and inducts on the actual local iterate Equation 1. This also establishes independence from the initial configuration outside the window.

For evaluation in parallel, evaluate the guards at every site and compute all prefix weights w_k and products e_k. The weight circuit combines carry-save pairs along a balanced tree and completes the binary sum at the end. For $w_k > 0$, the parity test for $M(w_k, T-k)$ checks whether every set bit of $T-k$ is also set in $w_k + T - k - 1$. The $w_k = 0$ branch uses the explicit zero-part convention. In the uniform construction the test uses a logspace-generated table of coefficients indexed by the possible prefix weights, together with equality and selection circuits. A balanced parity reduction combines the data contributions. The balanced arithmetic and selection stages give logarithmic depth. The logspace generator writes a polynomial-length code and the number of gates is bounded by its length.

2.2 Depth classification

Write $p(T)$ for the number of ones in the binary expansion of T, and d_s, d_h for the depths of the fixed guard circuits. A guard varies if it takes both Boolean values. All the bounds below are finite, including $T = 0$.

Guard condition	Lower bound	Uniform upper bound
$h \equiv 0$	0	$1 + d_s$
$h \equiv 1$, $s \equiv 0$	0	0
h varies, or $h \equiv 1$ and s varies	$\ell(T+1)$	$104(\ell(T+3)+1) + \max(d_s, d_h)$
$s \equiv h \equiv 1$	$p(T)$	$3p(T)$

Table 1: Depth bounds at legal lengths for fixed frozen guards.

Theorem 2 (uniform classification). For each fixed pair of guard circuits there is one logspace-uniform, polynomial-size De Morgan family that decides the whole language and achieves the appropriate upper bound in Table 1 at every legal length. It has depth $O(\log(n+2))$ at all lengths. The lower column bounds $D(T)$ and holds even for arbitrary fan-in-two Boolean gates. The language is in logspace-uniform NC^0 exactly when $h \equiv 0$, or when $h \equiv 1$ and $s \equiv 0$.

For $h \equiv 0$ and $T > 0$, the answer is $s(a_0)x_0$ by idempotence. The pure shift case returns x_T. At $T = 0$ every rule returns x_0.

When h varies, set all data bits except x_T to zero. The surviving term in Equation 3 is $x_T \prod_{j=0}^{T-1} h(a_j)$, regardless of s. A nonconstant Boolean guard has an essential bit, as seen along a single-bit path between two words with different guard values. Each of the first T control words contributes one essential bit on this restriction, and x_T contributes one more. When $h \equiv 1$ and s varies, fix $s(a_0) = 1$ and $s(a_i) = 0$

at sites 1 through T. Then $w_k = 1$ for every k and Equation 3 is parity of all $T + 1$ data bits. Both cases give the same lower bound by essential coordinates.

For $s \equiv h \equiv 1$, the coefficients are $\binom{T}{k} \bmod 2$. Over $\mathbb{F}_2$, the identity

$$(1 + z)^T = \prod_{j:\mathrm{bit}_{j(T)}=1} \left(1 + z^{2^j}\right) \tag{4}$$

shows that precisely the binary submasks of T occur. Their $2^{p(T)}$ data bits are essential. The logarithmic lower bound is therefore $p(T)$. A balanced XOR tree reduces those leaves in $p(T)$ rounds. Each XOR can be implemented with at most three De Morgan layers. The uniform generator scans $k = 0, \ldots, T$, emits the submask wires, and serializes a regular XOR reducer with $5p(T)2^{p(T)}$ gates. The depth is at most three when $T = 2^r$ and at least r when $T = 2^r - 1$. In particular the depth drops from at least r to at most three across these consecutive times for $r > 3$.

The executable classifier checks constancy of the guards on their finite alphabet. It visits control words by prefix and can stop a constancy test on its first counterexample, without allocating the alphabet as a list. An exhaustive test can still require 2^q words. For uniform prediction the guard circuits and the selected branch are fixed, and the transducer is uniform in the window length. At legal lengths its output is the selected predictor; at malformed lengths it writes a rejecting circuit. The theorem packages semantics, generation, size, and depth for that same family.

3 Transported controls

The next model gives the control track its own dynamics. Fix velocities $u, v \in \{0, 1\}$ and a Boolean control and data track (c_i, d_i). Define the synchronous rule

$$c_i' = c_{i+u}, \quad d_i' = c_i \wedge d_{i+v}. \tag{5}$$

The window has $2(T + 1)$ bits. With a single backwards data path, the data at time T equals

$$d_{i+vT} \wedge \bigwedge_{k=0}^{T-1} c_{i+kv+(T-1-k)u}, \tag{6}$$

where the conjunction is empty at $T = 0$. At the k-th path step we read control at site $i + kv$ and time $T - 1 - k$. The control has shifted by $(T - 1 - k)u$ initial sites, which yields Equation 6.

> **Theorem 3 (exact depth under transport).** Depth is zero at $T = 0$. For $T > 0$, it is exactly one if $u = v$ and exactly $\ell(T + 1)$ if $u \neq v$. The formal upper-bound construction uses a padded AND tree with at most $2m$ binary AND gates for m essential inputs. The lower bound applies to all fan-in-two Boolean circuits predicting the data bit.

For equal velocities all control indices coincide and idempotence leaves two essential inputs, one control and one data bit. For unequal velocities the initial control indices are $i, \ldots, i + T - 1$ in forward or reverse order. Together with the terminal data bit they are $T + 1$ essential coordinates. A balanced AND tree attains the depth bound for each legal window. This theorem is a per-time statement about circuits on those windows.

The pair $(u, v) = (0, 0)$ is the frozen no-hop case with stay guard c_i. Changing only u to one gives logarithmic depth although the data never hops. For $(u, v) = (1, 1)$ the data and control both move, yet their equal velocity gives constant depth. For this single-path rule, the relevant count is the number of distinct initial controls encountered in the moving frame of the data path. The time-dependent framework in [1] accommodates this rule. Its single path makes the exact depth accessible.

4 Relation to earlier work

Pascal parity and fast prediction. The odd-entry count $2^{p(T)}$ is classical. Rowland attributes the binary count to Glaisher and derives it from Lucas's theorem [2]. The fast cellular-automaton prediction work in [1] already uses the Pascal coefficients for homogeneous additive rules. Its modular-sum gates can have unbounded fan-in. The Pascal claim here is a per-time bounded-fan-in lower bound and a matching profile up to a constant factor with a verified uniform De Morgan generator.

Dependent linear tracks and the coefficient formula. Moore's general divide-and-conquer method for time- and space-dependent linear stages gives an $O(\log^2(T+2))$ bounded-fan-in prediction bound in the stated setting [1]. Freezing the controls and using the one-sided stay/hop stencil lets us use prefix statistics and reach $O(\log(T+2))$ in this specialization. The coefficient in Equation 3 also has a classical matrix analogue. The one-step data map is upper bidiagonal on the relevant finite window. Higham's divided-difference formula for functions of such matrices gives the product of intervening off-diagonal entries times a divided difference in the diagonal entries [3]. For the monomial z^T this divided difference is a complete homogeneous polynomial of degree $T-k$. On zero-one diagonal entries it counts the same weak compositions. The polynomial identity can be specialized to integer zero-one entries before reducing modulo two, with no division in $\mathbb{F}_2$.

Mechanization and the meaning of frozen. Silváši and Tomášek formalized bounded grids and computable cellular automata in Lean [4]. Their work supplies precedent for machine-checked cellular automata. The present artifact checks a rule-family depth classification and the actual circuit generator. The freezing model of Ollinger and Theyssier imposes a monotone order on each whole cell state [5]. In Equation 1 only the control word is fixed, and the data at a site can cycle. The comparison identifies antecedents for the main ingredients. Priority for the exact four-way classification and for the transport specialization remains undetermined.

5 Scope of the certificates

The frozen classification applies to an explicitly supplied cone and a fixed finite guard pair. Its logarithmic depth is measured against n, which grows linearly with T. Giving a succinct initial state and a binary time query defines a different problem. The theorem also assumes arbitrary wiring. With unbounded fan-in, the transported-control AND can be evaluated in one layer, and the counting argument no longer gives its logarithmic depth. Conversely, if each circuit layer has spatial wiring radius at most r, an essential coordinate at distance T forces $T \le dr$. This additional hypothesis is used for the repository's geometric lower bound in the OR automaton. For unrestricted wiring and fan-in two, that separate example has exact depth $\ell(2T+1)$.

Composition certificate. For an evolution F, a bit-string decoder I into its state space, and an observation O, suppose finite-register Boolean circuits implement initialization, composition along the executed path, and observation. If their depths are $\eta, \delta, \varepsilon$, repeated squaring produces $O\big(F^{T(I(x))}\big)$ from x with depth at most $\eta + 2\ell(T+1)\delta + \varepsilon$. A separate uniform compiler requires logspace generators for the primitives and charges copying during normalization. Its depth bound is $\eta + 1 + 2\ell(T+1)(\delta + 1) + \varepsilon$, with δ bounding both composition primitives. Membership in NC requires a polylogarithmic bound in the actual input length.

Simulation certificate. For a specified system, a logspace transducer from circuit value instances to complete queries $1^T 0x$, together with an observation-equality proof, makes its prediction language P-hard under logspace many-one reductions [6]. A polynomial-time decider for the target yields P-completeness. Assuming $\mathrm{P} \neq$ logspace-uniform NC, a verified hardness witness excludes the target from that parallel class. This conclusion gives no specific numerical depth rate. Rule 110 is a separate concrete P-completeness result in the literature [7].

6 Lean statement map and reproduction

The paths below are relative to `PredictionDepth/`. The names identify the machine-checked statements underlying the main claims.

Result	Lean module and declaration
Frozen iterate	`FrozenPascal/Semantics.lean` `iterate_data`

Uniform classification	`FrozenPascal/UniformClassification.lean`
	`uniform_depth_classification`
	`predictionLanguage_mem_uniformNC_zero_iff`

Pascal support	`FrozenPascal/HomogeneousPascal.lean` `pascalSupport_card`

Moving controls	`TransportedControl.lean`
	`iterate_data_true_iff`, `essential_prediction_iff`
	`exact_prediction_depth`, `constant_depth_iff`

Composition criterion	`EncodedPathComposition.lean`
	`EncodedPathImplementation.predictor_depth_le`
	`Complexity/UniformComposition.lean`
	`UniformRegisterWitness.depth_le_costs`

Simulation criterion	`Complexity/StandardHardness.lean`
	`SimulationWitness.prediction_P_complete`
	`Complexity/NCClosure.lean`
	`SimulationWitness.prediction_not_uniformNC`

The Lean toolchain and dependencies are pinned by `lean-toolchain` and `lake-manifest.json`. Run `./scripts/check.sh` at the repository root. The script checks entry-point coverage, builds with warnings treated as errors, audits transitive axiom dependencies of project declarations, and checks that the audit rejects an unapproved transitive axiom.

Bibliography

[1] C. Moore, "Predicting Nonlinear Cellular Automata Quickly by Decomposing Them into Linear Ones," *Physica D: Nonlinear Phenomena*, vol. 111, pp. 27–41, 1998, doi: 10.1016/S0167-2789(97)80003-6.

[2] E. Rowland, "The Number of Nonzero Binomial Coefficients Modulo p^a," *Journal of Combinatorics and Number Theory*, vol. 3, pp. 15–25, 2011, [Online]. Available: https://arxiv.org/abs/1001.1783

[3] N. J. Higham, "The Power of Bidiagonal Matrices," *Electronic Journal of Linear Algebra*, vol. 40, pp. 453–474, 2024, doi: 10.13001/ela.2024.8297.

[4] F. Silváši and M. Tomášek, "Lean Formalization of Bounded Grids and Computable Cellular Automata Defined Thereover," *Science of Computer Programming*, vol. 195, p. 102471, 2020, doi: 10.1016/j.scico.2020.102471.

[5] N. Ollinger and G. Theyssier, "Freezing, Bounded-Change and Convergent Cellular Automata," *Discrete Mathematics and Theoretical Computer Science*, vol. 24, no. 1, 2022, doi: 10.46298/dmtcs.5734.

[6] R. E. Ladner, "The Circuit Value Problem Is Log Space Complete for P," *ACM SIGACT News*, vol. 7, no. 1, pp. 18–20, 1975, doi: 10.1145/990518.990519.

[7] T. Neary and D. Woods, "P-Completeness of Cellular Automaton Rule 110," in *Automata, Languages and Programming (ICALP)*, in Lecture Notes in Computer Science, vol. 4051. 2006, pp. 132–143. doi: 10.1007/11786986_13.